

데이터 기반 방법을 통한 종속 확률 변수에 따른 강건 설계 최적화

류 희 창¹⁾

한양대학교 미래자동차공학과¹⁾

Robust Design Optimization under Dependent Random Variables by Data Driven Method

Hoecahng Ryu¹⁾

¹⁾Department of Automotive Engineering, Hanyang University, Seoul 135-080, Korea

Abstract : In this paper a new data-driven Robust Design Optimization method is proposed to overcome the limitations of traditional approaches based on strict assumptions about probability distributions. Conventional methods, such as generalized polynomial chaos expansion (GPCE), are based on explicit knowledge of the joint probability density function, which may not be available or is quite inaccurate for complex, dependent random variables. In this respect, I develop a framework for the direct estimation of the score function from data through a score matching algorithm. This approach allows for the accurate sensitivity analysis and efficient optimization with no presumed distribution forms, hence making it effective for solving RDO problems possessing arbitrary and datadriven uncertainties.

Key words : Robust design optimization(강건 설계 최적화), Dependent random variables(종속 확률 변수), Generalized polynomial chaos expansion(일반화된 다항식 카오스 전개), Score matching(점수 매칭), Data-driven method(데이터 기반 방법), Sensitivity analysis(민감도 해석)

Nomenclature

d : design vector
X : input random vector
Z : transformed random vector
 Ψ : multivariate orthonormal polynomial
C : expansion coefficient
 Θ : model parameter
 $L_{N,m}$: number of basis functions

Subscripts

d : design vector dependency
g : transformed mean vector dependency
k : index of design variable
l : index of performance function
m : order of polynomial approximation
 Θ : parameterized model

1. Introduction

Robust Design Optimizatio(RDO) is a systematic

methodology for engineering design that explicitly accounts for uncertainties in input variables, such as material properties, manufacturing tolerances, and operating conditions. Unlike traditional deterministic optimization, which often relies on conservative safety factors, RDO aims to find a design that is not only optimal in terms of performance but also insensitive (robust) to variations in uncertain parameters. This approach has been successfully applied across various engineering disciplines, including automotive, aerospace, and civil engineering, to manage risks associated with inherent uncertainties.

In practical applications, the objective and constraint functions are frequently evaluated using computationally expensive simulations, such as Finite Element Analysis(FEA). Consequently, efficient uncertainty quantification(UQ) methods are essential. While numerous surrogate methods like Polynomial Chaos Expansion(PCE) and Kriging have been

developed, most existing approaches assume statistical independence among input random variables. However, in reality, input variables often exhibit significant correlations or dependencies. Ignoring these dependencies can lead to inaccurate risk assessments and suboptimal or even unreliable designs. Therefore, advanced computational methods capable of handling arbitrary dependent probability distributions are required for effective RDO.

To address these challenges, this study proposes a new RDO methodology based on a data-driven approach. Although the existing Generalized Polynomial Chaos Expansion (GPCE) method offers the advantage of handling dependent variables, it is limited by the requirement of explicit knowledge of the probability density function (PDF) or the joint probability distribution. Therefore, this research introduces a Data-Driven GPCE framework that estimates the score function directly from data and performs sensitivity analysis based on it. This approach demonstrates that efficient and accurate robust design optimization can be achieved even in situations where the distribution information of input variables is incomplete or unknown.

2. Robust design optimization

This section establishes the mathematical framework for Robust design optimization (RDO). Consider a probability space (Ω_d, F_d, P_d) that depends on an M-dimensional design vector $\mathbf{d} \in D \subseteq \mathbb{R}^M$. The statistical uncertainties inherent in a complex mechanical system, such as loads, material properties, and geometry, are represented by an N-dimensional input random vector $\mathbf{X} = (X_1, \dots, X_N)^T$.

The probability law of $\mathbf{X}$ is fully defined by a family of joint probability density functions (PDF) $f_{\mathbf{x}}(\mathbf{x}; \mathbf{d})$, which are associated with the probability measures dependent on the design vector $\mathbf{d}$. While a design variable d_k can theoretically represent any distribution parameter, in this study, it is defined as the mean of the corresponding random variable X_k , consistent with standard engineering practices.¹⁾ (Lee and Rahman 2021)

2.1 Problem definition

Let $y_l(\mathbf{X})$ be the equation describing output functions of a complex system on measurable space (Ω_d, F_d) . Commonly referred to as response or performance functions in applications. In general case, it mostly depends on $\mathbf{d}$ via the probability law of $\mathbf{X}$, although not be explicitly represented. From the purpose of RDO, it is performed by minimizing the mean and standard deviation of the response function, can be formulated as following to

$$\begin{aligned} \min_{\mathbf{d} \in D \subseteq \mathbb{R}^M} & \{ \mathbf{E}_{\mathbf{d}}[y_0(\mathbf{X})], \sqrt{\text{var}_{\mathbf{d}}[y_0(\mathbf{X})]} \}, \\ \text{subject to } & a_l \sqrt{\text{var}_{\mathbf{d}}[y_l(\mathbf{X})]} - \mathbf{E}_{\mathbf{d}}[y_l(\mathbf{X})] \leq 0, \\ & l = 1, \dots, K, 1 \leq K < \infty \\ & d_{k,L} \leq d_k \leq d_{k,U}, k = 1, \dots, M \end{aligned}$$

where

$$\mathbf{E}_{\mathbf{d}}[y_l(\mathbf{X})] := \int_{\mathbf{A}^N} y_l(\mathbf{x}) f_{\mathbf{x}}(\mathbf{x}; \mathbf{d}) d\mathbf{x}$$

is the mean of $y_l(\mathbf{X})$ and

$$\text{var}_{\mathbf{d}}[y_l(\mathbf{x})] := \mathbf{E}_{\mathbf{d}}[y_l(\mathbf{X}) - \mathbf{E}_{\mathbf{d}}[y_l(\mathbf{X})]]^2$$

is the variance of $y_l(\mathbf{x})$. In this bi-objective optimization problem, the objective functions might have trade-off relationships; minimizing one could lead to increasing the other one and vice versa. It gives infinite number of optimal solutions, typically called Pareto optimal solutions.

In this paper, the weighted sum approach is used to handle with this problem, transforming the bi-objective optimization problem into single-objective optimization problem. It can be formulated as following - Problem formulation.

2.2 Problem formulation

As mentioned above, the optimization problem can be formulated to single-objective function using weighted sum approach as

$$\begin{aligned} \min_{\mathbf{d} \in D \subseteq \mathbb{R}^M} c_0(\mathbf{d}) & := G(\mathbf{E}_{\mathbf{d}}[y_0(\mathbf{X})], \sqrt{\text{var}_{\mathbf{d}}[y_0(\mathbf{X})]}), \\ & := w_1 \frac{\mathbf{E}_{\mathbf{d}}[y_0(\mathbf{X})]}{\mu_0^*} + w_2 \frac{\sqrt{\text{var}_{\mathbf{d}}[y_0(\mathbf{X})]}}{\sigma_0^*}, \\ \text{subject to } & a_l \sqrt{\text{var}_{\mathbf{d}}[y_l(\mathbf{X})]} - \mathbf{E}_{\mathbf{d}}[y_l(\mathbf{X})] \leq 0, \\ & l = 1, \dots, K, \\ & d_{k,L} \leq d_k \leq d_{k,U}, k = 1, \dots, M \end{aligned}$$

The two objectives (mean and variance) are normalized by their initial value and weighted according to the designer's purpose.

2.3 Alternative formulation

Since the design variables are the means of some input random variables, a linear transformation such as shifting and scaling could provide better approaches to handle with RDO problems. In this paper, scaling is employed exclusively to address more general problems that reflect real-world properties; Real-world data exhibits variance proportional to its mean, with higher mean values corresponding to higher variances.

scaling Let $\mathbf{Z} = (\mathbf{Z}_1, \mathbf{Z}_2, \dots, \mathbf{Z}_N)^T$ be an N -dimensional vector of new random variables obtained by scaling $\mathbf{X}$ as

$$\mathbf{Z} = \text{diag}[r_1, \dots, r_N] \mathbf{X}$$

where $\mathbf{r} = (r_1, r_2, \dots, r_N)^T$ is an N -dimensional vector of deterministic variables. The PDF of $\mathbf{Z}$ is

$$\begin{aligned} f_{\mathbf{Z}}(\mathbf{z}; \mathbf{d}) &= \mathbb{J} f_{\mathbf{X}}(\mathbf{x}; \mathbf{d}) = \left| \frac{1}{r_1 \dots r_N} \right| f_{\mathbf{X}}(\mathbf{x}; \mathbf{d}) \\ &= \left| \frac{1}{r_1 \dots r_N} \right| f_{\mathbf{X}}(\text{diag}[1/r_1, \dots, 1/r_N] \mathbf{z}; \mathbf{d}). \end{aligned}$$

Here, the new M -dimensional vector $\mathbf{g} = (g_1, \dots, g_M)^T$ can be defined as $g_k = \mathbf{E}[Z_{ik}]$, $k = 1, \dots, M$.

Once the new output function $h_l(\mathbf{Z}; \mathbf{r}) := y_l(\mathbf{X})$ for the transformed random variables $\mathbf{Z}$ is defined, the new RDO problem is formulated to

$$\begin{aligned} \min_{\mathbf{d} \in D \subseteq \mathbf{R}^M} c_0(\mathbf{d}) &:= G(\mathbf{E}_{\mathbf{g}(\mathbf{d})}[h_0(\mathbf{Z}; \mathbf{r})], \sqrt{\text{var}_{\mathbf{g}(\mathbf{d})}[h_0(\mathbf{Z}; \mathbf{r})]}), \\ \text{subject to } c_l(\mathbf{d}) &:= \alpha \sqrt{\text{var}_{\mathbf{g}(\mathbf{d})}[h_l(\mathbf{Z}; \mathbf{r})] - \mathbf{E}_{\mathbf{g}(\mathbf{d})}[h_l(\mathbf{Z}; \mathbf{r})]} \leq 0, \\ l &= 1, \dots, K, \\ d_{k,L} &\leq d_k \leq d_{k,U}, k = 1, \dots, M, \end{aligned}$$

where

$$\mathbf{E}_{\mathbf{g}(\mathbf{d})}[h_l(\mathbf{Z}; \mathbf{r})] := \int_{\mathbf{A}^N} h_l(\mathbf{z}; \mathbf{r}) f_{\mathbf{Z}}(\mathbf{z}; \mathbf{r}) d\mathbf{z}$$

is the mean of $h_l(\mathbf{Z}; \mathbf{r})$ and

$$\text{var}_{\mathbf{g}(\mathbf{d})}[h_l(\mathbf{Z}; \mathbf{r})] := \mathbf{E}_{\mathbf{g}(\mathbf{d})}[h_l(\mathbf{Z}; \mathbf{r}) - \mathbf{E}_{\mathbf{g}(\mathbf{d})}[h_l(\mathbf{Z}; \mathbf{r})]]^2$$

is the variance of $h_l(\mathbf{Z}; \mathbf{r})$. In accordance with the aforementioned methodology, the probability measure of $\mathbf{Z}$ is fixed during design iterations. Consequently, the necessity to recalculate measure-associated quantities is circumvented.

3. Design sensitivity analysis

Solving the RDO problem defined in the previous section using a gradient-based optimization algorithm requires the evaluation of not only the objective and constraint functions but also their sensitivities with respect to design variables. In this study, efficient Design Sensitivity Analysis is achieved by integrating

Generalized Polynomial Chaos Expansion (GPCE) with the Score Function method. This section outlines the theoretical foundation for this approach.

3.1 Generalized polynomial chaos expansion

Enter the contents. Enter the contents. Enter the contents. Enter the contents. Enter the contents. Enter the contents. Enter the contents.

The Generalized Polynomial Chaos Expansion (GPCE) is employed as a robust surrogate modeling technique to approximate the random output function $h(\mathbf{Z}; \mathbf{r})$ of complex systems. Distinct from conventional Polynomial Chaos Expansion (PCE) methods, which are typically limited to independent random variables^{2),3)} (Wiener 1938; Xiu and Karniadakis 2002), the GPCE framework utilized in this research is specifically formulated to accommodate input random variables governed by arbitrary, dependent probability distributions. In this framework, any square-integrable random variable $h(\mathbf{Z}; \mathbf{r}) \in L^2(\Omega_{\mathbf{d}}, F_{\mathbf{d}}, P_{\mathbf{d}})$ can be represented as a spectral expansion using multivariate orthonormal polynomials $\psi_i(\mathbf{Z}; \mathbf{g})$. This infinite series expansion is expressed as:

$$h(\mathbf{Z}; \mathbf{r}) \sim \sum_{i=1}^{\infty} C_i(\mathbf{r}) \psi_i(\mathbf{Z}; \mathbf{g})$$

where $C_i(\mathbf{r})$ denotes the expansion coefficients corresponding to the orthonormal basis $\psi_i(\mathbf{Z}; \mathbf{g})$. However, since the theoretical expansion involves an infinite number of terms, practical implementation necessitates truncation. By adopting a total-degree index set to retain polynomial terms up to a specified order m , the m -th order GPCE approximation is formulated as:

$$h_m(\mathbf{Z}; \mathbf{r}) \sim \sum_{i=1}^{L_{N,m}} C_i(\mathbf{r}) \psi_i(\mathbf{Z}; \mathbf{g})$$

where $L_{N,m}$ represents the finite number of basis functions retained in the approximation. The subsequent subsections provide detailed procedures for constructing these orthonormal basis functions and estimating the associated expansion coefficients.

3.2.1 Orthonormal polynomials

When input random variables $\mathbf{Z} = (\mathbf{Z}_1, \mathbf{Z}_2, \dots, \mathbf{Z}_N)^T$ are statistically dependent, their joint probability measure is generally not of product type. Consequently, multivariate orthonormal polynomials cannot be

constructed simply by tensor products of univariate polynomials. To address this, a three-step algorithm based on the whitening transformation is employed to generate a set of multivariate orthonormal polynomials $\psi_m(\mathbf{Z}; \mathbf{g})$ consistent with an arbitrary probability measure $f_z(\mathbf{z}, \mathbf{g})d\mathbf{z}$.

1) Monomial basis Construction: For a given maximum polynomial degree $m \in \mathbf{N}_0$, an $L_{N,m}$ -dimensional vector of monomials $\mathbf{P}_m(\mathbf{z})$ is created. This vector contains all monomials $\mathbf{z}^{\mathbf{j}}$ where the total degree $|\mathbf{j}| \leq m$, arranged according to a specific monomial order (e.g. graded lexicographic order).

2) Moment matrix calculation: The monomial moment matrix $\mathbf{G}_m$ of size $L_{N,m} \times L_{N,m}$ is constructed by calculating the expectation of the outer product of the monomial vector:

$$\mathbf{G}_m := \mathbf{E}_{\mathbf{g}}[\mathbf{P}_m(\mathbf{Z})\mathbf{P}_m^T(\mathbf{Z})] = \int \mathbf{P}_m(\mathbf{z})\mathbf{P}_m^T(\mathbf{z})f_z(\mathbf{z}; \mathbf{g})d\mathbf{z}$$

This matrix captures the statistical moments of the input variables.

3) Whitening transformation: An $L_{N,m} \times L_{N,m}$ whitening matrix $\mathbf{W}_m$ is determined, typically via Cholesky decomposition of the moment matrix $\mathbf{G}_m$, such that $\mathbf{W}_m^T \mathbf{W}_m = \mathbf{G}_m^{-1}$. The multivariate orthonormal polynomial basis $\psi_i(\mathbf{Z}; \mathbf{g})$ is then generated by applying this linear transformation to the monomial vector:

$$\psi_i(\mathbf{Z}; \mathbf{g}) = \mathbf{W}_m \mathbf{P}_m(\mathbf{z})$$

This procedure ensures that the resulting polynomial basis functions satisfy the orthonormality condition $\mathbf{E}_{\mathbf{g}}[\psi_i(\mathbf{z}; \mathbf{g})\psi_j(\mathbf{z}; \mathbf{g})] = \delta_{ij}$, which is essential for the subsequent generalized polynomial chaos expansion.

3.2.2 Expansion coefficients

The expansion coefficients of the GPCE approximation, $C_i(\mathbf{r})$, are theoretically defined by high-dimensional integrals involving the product of the output function and the orthonormal polynomials. For arbitrary probability distributions and high-dimensional problems, exact evaluation of these integrals is computationally prohibitive. To address this, a regression-based approach, specifically Standard Least-Squares (SLS), is employed to estimate the coefficients efficiently.

Given an input-output dataset $\{\mathbf{z}^{(l)}, h(\mathbf{z}^{(l)}; \mathbf{r})\}_{l=1}^L$ of size L , generated via sampling methods such as

Quasi-Monte Carlo Simulation (QMCS), the approximate expansion coefficients $\tilde{\mathbf{c}} = (\tilde{C}_1, \dots, \tilde{C}_{L_{N,m}})^T$ are determined by solving the linear system:

$$\mathbf{A}\mathbf{c} = \mathbf{b}$$

where $\mathbf{A}$ is an $L \times L_{N,m}$ matrix containing the evaluations of the orthonormal polynomials $\tilde{\psi}_i(\mathbf{z}^{(l)}; \mathbf{g})$ at the sample points, and $\mathbf{b}$ is an L -dimensional vector of the corresponding output function values $h(\mathbf{z}^{(l)}; \mathbf{r})$. The SLS method estimates the coefficients by minimizing the sum of squared residuals between the actual function values and the GPCE approximation:

$$\hat{\mathbf{c}}_m := \frac{1}{L} \sum_{l=1}^L \left[h(\mathbf{z}^{(l)}; \mathbf{r}) - \sum_{i=1}^{L_{N,m}} \tilde{C}_i \tilde{\psi}_i(\mathbf{z}^{(l)}; \mathbf{g}) \right]^2$$

The optimal coefficient vector $\hat{\mathbf{c}}$ is then obtained by solving the normal equations:

$$\hat{\mathbf{c}} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b}$$

3.2 Statistical moment

Using the GPCE coefficient above, the statistical moments - mean and variance - can be calculated as follows.

mean

$$\mathbf{E}_{\mathbf{g}}[h_m(\mathbf{Z}; \mathbf{r})] = C_1(\mathbf{r})$$

variance

$$\text{var}_{\mathbf{g}}[h_m(\mathbf{Z}; \mathbf{r})] = \sum_{i=2}^{L_{N,m}} C_i^2(\mathbf{r})$$

3.3 Score function

To solve the RDO sub-problems using gradient-based methods, it is necessary to calculate the sensitivities of the first two statistical moments (where $r = 1, 2$) with respect to the design variable d_k . By applying the differential operator to the expectation integral and utilizing the log-derivative method, the first-order score function $s_k(\mathbf{Z}; \mathbf{g})$ is defined as

$$s_k(\mathbf{Z}; \mathbf{g}) := \frac{\partial \ln f_z(\mathbf{Z}; \mathbf{g})}{\partial g_k}$$

This function represents the sensitivity of the log-probability density with respect to the distribution parameters. Its primary utility lies in transforming the derivative of an expectation into the expectation of a product. Consequently, the design sensitivity of the moments can be expressed as

$$\frac{\partial \mathbf{E}_{\mathbf{g}}[h'(\mathbf{Z}; \mathbf{r})]}{\partial d_k} = \frac{\partial g_k}{\partial d_k} \mathbf{E}_{\mathbf{g}}[h'(\mathbf{Z}; \mathbf{r}) s_k(\mathbf{Z}; \mathbf{g})]$$

This formulation is essential because it enables the simultaneous evaluation of both the statistical moments and their design sensitivities within a single stochastic simulation, thereby avoiding separate and costly perturbation analyses.

3.4 Approximate sensitivities

Since the exact sensitivity analysis involves infinite basis functions, a practical implementation requires truncating the Generalized Polynomial Chaos Expansion (GPCE). By approximating the response function $h(\mathbf{Z}; \mathbf{r})$ with an m -th order expansion and the score function $s_k(\mathbf{Z}; \mathbf{g})$ with an m' -th order expansion, the sensitivity of the first moment(mean) and second moment(variance) values can be computed numerically.

The approximate sensitivity of the mean with respect to the design variable d_k is derived as the inner product of the expansion coefficients

$$\frac{\partial \mathbf{E}_{\mathbf{g}}[h_m(\mathbf{Z}; \mathbf{r})]}{\partial d_k} = \frac{\partial g_k}{\partial d_k} \sum_{i=2}^{L_{\min}} C_i(\mathbf{r}) D_{k,i}(\mathbf{g})$$

where $C_i(\mathbf{r})$ and $D_{k,i}(\mathbf{g})$ are the GPCE coefficients of the response and the score function, respectively. The summation limit is defined as $L_{\min} := \min(L_{N,m}, L_{N,m'})$, and the sum begins at $i=2$ because the expectation (first coefficient) of the score function is zero. The square term for $h(\mathbf{X})$ can also be handled in the same way as previously described.

4. Score function estimation algorithm

In the context of sensitivity analysis, it is imperative to ascertain the precise value of the score function term to facilitate the calculation of sensitivities during the optimization process. In this paper, the score matching algorithm is introduced as a means of estimating the score function for an arbitrary data set of unknown probability distribution shape.

4.1 Score matching algorithm

Let $\mathbf{X} = (X_1, X_2, \dots, X_N)$ be N -dimensional data set and $\mathbf{S} = (S_1, S_2, \dots, S_N)$ be a score function matching to each data point. In this context, the score function is defined as the gradient vector of the logarithm of the probability density function, with respect to $\mathbf{X}$. Let $f_{\mathbf{X}}$ be the real value function on $\mathbf{X} \in \mathbf{R}^D$. Based on the

energy model, the probability model is defined as

$$p_{\theta}(\mathbf{x}) = \frac{e^{-f_{\theta}(\mathbf{x})}}{Z_{\theta}},$$

where $Z_{\theta} = \int e^{-f_{\theta}(\mathbf{x})}$,

the unknown normalizing constant. In this context, the probability model is unnormalized. Score matching algorithm bypasses the intractable Z_{θ} by only considering the scores, which is defined as the gradient of the log-density w.r.t. the random variable.⁴⁾ (Y.Song 2024) So, the score of $p_{\theta}(\mathbf{x})$ is

$$\nabla_{\mathbf{x}} \log p_{\theta}(\mathbf{x}) = -\nabla_{\mathbf{x}} f_{\theta}(\mathbf{x}).$$

Now, the score matching use following objective to estimate score function to

$$\min \frac{1}{2} \mathbf{E}_{p_{\text{data}}} [\|\nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\theta}(\mathbf{x})\|^2]$$

which is widely known as the Fisher divergence where $p_{\text{data}}(\mathbf{x})$ is the data distribution. But still, $\nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$, the score of the data distribution is also unknown. To train the score model, by applying partial integration to above function, it can be deformed as

$$\mathbf{E}_{p_{\text{data}}} \left[\text{tr}(\nabla_{\mathbf{x}}^2 \log p_{\theta}(\mathbf{x})) = \frac{1}{2} \nabla \cdot \|\nabla_{\mathbf{x}} \log p_{\theta}(\mathbf{x})\|^2 \right] + \text{const},$$

where $\nabla_{\mathbf{x}}^2$ denotes the Hessian with respect to $\mathbf{x}$. This object is characterized by its exclusive reliance on the function of $\nabla_{\mathbf{x}} \log p_{\theta}(\mathbf{x})$, rendering it particularly well-suited for the purpose of learning the score function.

4.1.1 Sliced score matching

In the context of high-dimensional data sets, the calculation of the Hessian of the probability density function can present significant challenges. Sliced Score Matching (SSM) is an algorithm that decomposes data dimensions to facilitate the calculation of scores, thereby obviating the necessity of performing costly computations of the Hessian matrix. The main idea of sliced score matching is this; the two vector fields are equivalent if and only if their scalar fields corresponding to projections onto all directions are the same.⁵⁾ (Y.Song 2024) The random projected objected Fisher divergence is

$$\mathbf{E}_{p_{\text{data}}} \left[\mathbf{v}^T \nabla_{\mathbf{x}}^2 \log p_{\theta}(\mathbf{x}) \mathbf{v} + \frac{1}{2} (\mathbf{v}^T \nabla_{\mathbf{x}} \log p_{\theta}(\mathbf{x}))^2 \right] + \text{const}, \quad (1)$$

where $\mathbf{v}$ is denoted as the random projection direction. Here, $\mathbf{v}^T \nabla_{\mathbf{x}} \log p_{\theta}(\mathbf{x})$ can be computed by simple backpropagation with $O(1)$ time complexity, consequently reducing the computational cost. For the

data set $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\}$, equation (1) can be generalized to

$$\frac{1}{NM} \sum_{i=1}^N \sum_{j=1}^M \left[\mathbf{v}_{ij}^T \nabla_{\mathbf{x}}^2 \log p_{\theta}(\mathbf{x}_i) \mathbf{v}_{ij} + \frac{1}{2} (\mathbf{v}_{ij}^T \nabla_{\mathbf{x}} \log p_{\theta}(\mathbf{x}_i))^2 \right],$$

where M is denoted as the number of random projection directions $\{\mathbf{v}_{11}, \mathbf{v}_{12}, \dots, \mathbf{v}_{iM}\}$.

4.1.2 Denoising Score Matching

What distinguishes denoising score matching (DSM) from classical score matching is how to handle the challenge of estimating scores directly. Instead of trying to estimate the score of the original data distribution, DSM works with a noised version of the data. This approach avoids the need to compute Hessian term, reducing the computational cost for high-dimensional data. The main procedure of the algorithm is as follows. First, gradually add gaussian noise to original data to generate a new data set

$$\tilde{\mathbf{x}} = \mathbf{x} + \epsilon$$

with smooth probability distribution. Then the marginal probability distribution of the new data set $\tilde{\mathbf{x}}$ follows

$$p_{\sigma}(\tilde{\mathbf{x}}) = \int p_{\text{data}}(\mathbf{x}) N(\tilde{\mathbf{x}}|\mathbf{x}, \sigma^2 \mathbf{I}) d\mathbf{x}$$

where the score function is denoted as

$$S_{\sigma}(\tilde{\mathbf{x}}) = \nabla_{\tilde{\mathbf{x}}} \log p_{\sigma}(\tilde{\mathbf{x}}).$$

The objective of DSM is to learn the model $s_{\theta}(\tilde{\mathbf{x}})$ which approximate the score of noised data $s_{\sigma}(\tilde{\mathbf{x}})$. Thus the loss function for the learning is defined as

$$\mathbf{E}_{\tilde{\mathbf{x}} \sim p_{\sigma}} \| s_{\theta}(\tilde{\mathbf{x}}) - \nabla_{\tilde{\mathbf{x}}} \log p_{\sigma}(\tilde{\mathbf{x}}) \|^2,$$

and it has been proved to be the same parameter θ that makes above loss function to be minimized when

$$\mathbf{E}_{\tilde{\mathbf{x}} \sim p_{\text{data}} \epsilon \sim N} \left\| s_{\theta}(\mathbf{x} + \epsilon) + \frac{\epsilon}{\sigma^2} \right\|^2 + \text{const}$$

is either minimized under parameter θ . Now, when the noise is sufficiently small, the approximated score of noisy data converges to that of the original data. In the ‘denoising process’, from the high level of noised data, the estimated score offer an information of direction to a lower level of noised data. This is commonly known as ‘Langevin dynamics’. Following this process, the score function of the original data can be obtained.

4.2 Score function for Design variables

So far, the score function estimated by score

matching algorithm is the score with respect to the probability variable ‘ $\mathbf{x}$ ’. However what is needed to sensitivity analysis is the score function with respect to the design variable $\mathbf{d}$ which is the mean of probability variables $\mathbf{x}$ for each dimension of data. The score function with respect to design variable $\mathbf{d}$ can be calculated by following procedure.

1) $\mathbf{Z} \rightarrow \mathbf{X}$

Let $\mathbf{Z} = (\mathbf{Z}_1, \mathbf{Z}_2, \dots, \mathbf{Z}_N)^T$ be an N -dimensional vector of new random variables obtained by scaling $\mathbf{X}$ as

$$\mathbf{Z} = \text{diag}[r_1, \dots, r_N] \mathbf{X}$$

where $\mathbf{r} = (r_1, r_2, \dots, r_N)^T$ is an N -dimensional vector of deterministic variables. The PDF of $\mathbf{Z}$ is

$$\begin{aligned} f_{\mathbf{Z}}(\mathbf{z}; \mathbf{d}) &= \mathbb{J} f_{\mathbf{X}}(\mathbf{x}; \mathbf{d}) = \left| \frac{1}{r_1 \dots r_N} \right| f_{\mathbf{X}}(\mathbf{x}; \mathbf{d}) \\ &= \left| \frac{1}{r_1 \dots r_N} \right| f_{\mathbf{X}}(\text{diag}[1/r_1, \dots, 1/r_N] \mathbf{z}; \mathbf{d}). \end{aligned}$$

From above equation,

$$\begin{aligned} \log f_{\mathbf{Z}}(\mathbf{z}; \mathbf{d}) &= \log f_{\mathbf{X}}(\mathbf{x}; \mathbf{d}) - \sum_{i=1}^N \log r_i \\ \nabla_{\mathbf{z}} \log f_{\mathbf{Z}}(\mathbf{z}; \mathbf{d}) &= \nabla_{\mathbf{x}} \log f_{\mathbf{X}}(\mathbf{x}; \mathbf{d}). \end{aligned} \quad (2)$$

Applying the chain rule,

$$\frac{\partial \mathbf{z}}{\partial \mathbf{x}} \cdot \nabla_{\mathbf{z}} \log f_{\mathbf{Z}}(\mathbf{z}; \mathbf{d}) = \nabla_{\mathbf{x}} \log f_{\mathbf{X}}(\mathbf{x}; \mathbf{d}),$$

$$\nabla_{\mathbf{x}} \log f_{\mathbf{X}}(\mathbf{x}; \mathbf{d}) = \text{diag}[r_1, r_2, \dots, r_N] \nabla_{\mathbf{z}} \log f_{\mathbf{Z}}(\mathbf{z}; \mathbf{d}).$$

2) $\mathbf{X} \rightarrow \mathbf{d}$

① Score function w.r.t. $\mathbf{x}$

from (2), applying chain rule $u_i = x_i/d_i$ to be

$$\begin{aligned} S_{\mathbf{x}}(\mathbf{x})_i &= \frac{\partial \log f_{\mathbf{Z}}}{\partial u_i} \cdot \frac{\partial u_i}{\partial x_i} = (\nabla_u \log p_{\mathbf{Z}})_i \cdot \frac{1}{d_i}, \\ (\nabla_u \log p_{\mathbf{Z}})_i &= S_{\mathbf{x}}(\mathbf{x})_i \cdot d_i. \end{aligned} \quad (3)$$

② Score function w.r.t. design variable $\mathbf{d}$

$$S_{\mathbf{d}}(\mathbf{x})_i = \frac{\partial}{\partial d_i} \left(\log f_{\mathbf{Z}}(\mathbf{x}/\mathbf{d}) - \sum_k \log d_k \right)$$

In this equation the first term

$$\frac{\partial}{\partial d_i} \log p_{\mathbf{Z}}(\mathbf{x}/\mathbf{d}) = (\nabla_u \log p_{\mathbf{Z}})_i \cdot \frac{\partial (x_i/d_i)}{\partial d_i}, \quad (5)$$

and the second term

$$-\frac{\partial}{\partial d_i} \left(-\sum_k \log d_k \right). \quad (6)$$

Now, insert (3) into (5) and combine with (6), then

$$s_{\mathbf{d}}(\mathbf{x})_i = -\frac{1}{d_i} \cdot (1 + x_i s_{\mathbf{x}}(\mathbf{x})_i).$$

The combination of the first and second elements, with the result denoted by matrix notation, yields the final equation as follows:

$$S_k(\mathbf{X}; \mathbf{d}) = -\frac{1}{\mathbf{d}_k} (1 + \mathbf{r}_k \times S_k(\mathbf{Z}; \mathbf{g}) \odot X_k).$$

transformation equation'.

This equation will be referred as the 'score space

Table 1 The list of parameters, GPCE orders and sample sizes (Example 1,2)

	m		m'	L		L'	L''
Example1	y_0	y_1	s_k	y_0	y_1	s_k	
	4	1	1	45	9	30	1×10^6
Example 2	y_l		s_k	y_l		s_k	
	2		3	198		2860	2×10^6

6. Practical Example

6.1 Example 1: Optimisation of a mathematical function

The purpose of solving following mathematical problem is to prove the accuracy of score estimation algorithm. In this mathematical problem, I focused on how much the estimated score function is fitted to theoretical score function as well as the optimization result.

There is a two-dimensional Gaussian random vecotor $\mathbf{X} = (X_1, X_2)^T$ with dependent components, which have means $E[X_1] = d_1$ and $E[X_2] = d_2$. Given the design vector $\mathbf{d} = (d_1, d_2)^T$, the objective of this example is to

$$\begin{aligned} \min_{\mathbf{d} \in D} c_0(\mathbf{d}) &:= \frac{\sqrt{\text{var}_{\mathbf{d}}[y_0(\mathbf{X})]}}{\sigma_0^*}, \\ \text{subject to } c_1(\mathbf{d}) &:= 3\sqrt{\text{var}_{\mathbf{d}}[y_1(\mathbf{X})]} - E_{\mathbf{d}}[y_1(\mathbf{X})] \leq 0, \\ &0 \leq d_1 \leq 10, 0 \leq d_2 \leq 10, \end{aligned}$$

$$\text{where } y_0(\mathbf{X}) = (X_1 - 4)^3 + (X_1 - 3)^4 + (X_2 - 5)^2 + 10,$$

$$\text{and } y_1(\mathbf{X}) = X_1 + X_2 - 6.45$$

are two random output functions of $\mathbf{X}$. The initial design vector $\mathbf{d}_0$. The approximate optimal solution is denoted by $\tilde{\mathbf{d}}^*$. The standard deviations of X_1 and X_2 are $0.15d_1$ and $0.15d_2$, respectively. The correlation coefficient between X_1 and X_2 is - 0.5. The normalizing factor $\sigma_0^* = 45$.

To solve above problem, the sliced score matching method is employed to predict the score function with a high degree of accuracy in low-dimensional problems. Formerly studied by Lee & Rahman (2021) for correlated Gaussian variables, the second moment properties and score functions i.e. sensitivities are estimated with the assumption of Gaussian distribution.⁶⁾ In this example, the procedure is executed through data driven method.

Table 2 Optimization results of mathematical formulations (Example 1)

Results	Direct GPCE	Exact
d_1^*	3.3876	3.3906
d_2^*	5.0707	5.0673
$c_0(\tilde{\mathbf{d}}^*)$	0.0682	0.0682
$c_1(\tilde{\mathbf{d}}^*)$	-0.0003	-6.6613×10^{-15}
$\sqrt{\text{var}_{\mathbf{d}}[y_{\tilde{\mathbf{d}}^*}(\mathbf{X})]}$	1.1591	1.1592

6.2 Example2: Optimal sizing design of a ten-bar truss structures

In the second example, a linear-elastic ten-bar truss (cf. Lee and Rahman, 2021) was studied to evaluate the data-driven GPCE methods.⁷⁾ As shown in Fig.1, the initial conditions for truss structure are identical to those observed in previous studies. The key difference between this example and previous studies is that the conditions for the random variables are only specified at the initial stage for generating data sets. During the iteration, these conditions are not used to calculate

GPCE.

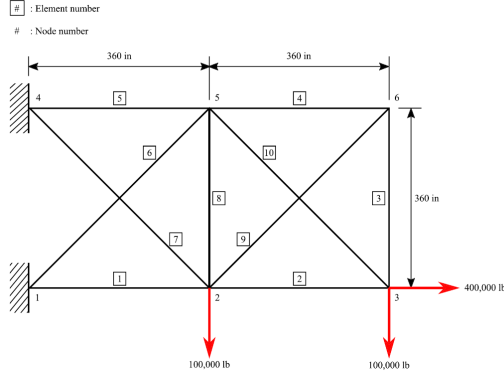


Fig. 1 A ten-bar truss structure

$$\min_{\mathbf{d} \in D \subseteq \mathbf{R}^M} c_0(\mathbf{d}) := 0.5 \frac{\mathbf{E}_{\mathbf{d}}[y_0(\mathbf{X})]}{6,285} + 0.5 \frac{\sqrt{\text{var}_{\mathbf{d}}[y_0(\mathbf{X})]}}{192},$$

$$\text{subject to } c_l(\mathbf{d}) := 3\sqrt{\text{var}_{\mathbf{d}}[y_l(\mathbf{X})]} - \mathbf{E}_{\mathbf{d}}[y_l(\mathbf{X})] \leq 0,$$

$$l = 1, \dots, 11,$$

$$0.1 \text{ in}^2 \leq d_k \leq 30 \text{ in}^2, k = 1, \dots, 10,$$

where

$$y_0(\mathbf{X}) = 0.1 \sum_{i=1}^{10} l_i X_i$$

is the random mass of the truss with l_i , $i = 1, \dots, 10$,

Table 3 Optimization result of ten-truss(Example 2)

	Gaussian based GPCE	Data driven GPCE	Theoretical value
d_1^*, in^2	5.0210	4.3854	4.6435
d_2^*, in^2	13.8747	13.3245	13.9252
d_3^*, in^2	0.1292	0.1316	0.1000
d_4^*, in^2	0.2136	0.1000	0.1000
d_5^*, in^2	6.1373	4.9531	4.6433
d_6^*, in^2	0.2793	0.1595	0.1000
d_7^*, in^2	0.2793	13.1483	12.9824
d_8^*, in^2	5.0234	4.4857	4.4740
d_9^*, in^2	0.1009	0.1127	0.1385
d_{10}^*, in^2	4.6974	3.2382	0.1385
$c_0(\mathbf{d}^*)$	0.345412	0.315209	0.31824

7. Conclusion

This study proposed and validated a new data-driven framework for Robust Design Optimization that can handle dependent random variables without explicit knowledge of their joint probability distributions. The major conclusions and the contributions of this research are summarized below:

1) The score matching algorithm was used to verify the performance of data-driven score prediction. It can be observed from the results that the proposed method was able to estimate the score functions from data samples

representing bar lengths and

$$y_l(\mathbf{X}) = \begin{cases} 25,000 - \sigma_l(\mathbf{X}), & \text{if } l = 1, \dots, 9, \\ 75,000 - \sigma_l(\mathbf{X}), & \text{if } l = 10, \\ 5 - v_2(\mathbf{X}), & \text{if } l = 11, \end{cases}$$

are 11 stochastic performance functions. The initial design is $\mathbf{d}_0 = (15, \dots, 15)^T \text{ in}^2$. The approximate optimal solution is denoted by $\tilde{\mathbf{d}}^* = (\tilde{d}_1^*, \dots, \tilde{d}_{10}^*)^T$.

Table.1 summarises the results of the design optimization in terms of the comparison between the Gaussian-based GPCE, the data-driven GPCE and the theoretical value. These results show that the data-driven method provides much better insights for handling unknown data. For the lognormal data distribution with low variance, the shape seems to be very similar with Gaussian distribution making it hard to predict the type of data distribution. Consequently, leading to the wrong point being selected for the optimal result. As demonstrated in the result, the data-driven method-based GPCE method exhibits reasonable performance in RDO for unknown data.

rather accurately. This effectively avoids the limitation of traditional methods, in which known probability density functions are relied upon, thereby allowing for precise sensitivity analysis even when distribution information might be incomplete.

2) A Data-Driven Generalized Polynomial Chaos Expansion surrogate model was developed that had universal applicability. This framework allows uncertainty quantification in complex systems in an efficient way by integrating estimated scores into the sensitivity analysis. In contrast to other surrogate models, which are restricted to specific distributions or

independent variables, this data-driven GPCE was able to model arbitrary dependent input uncertainties. ,2024.

3) the applicability of the developed surrogate model to real-world RDO problems was verified by applying it to the classic benchmark problem of a ten-bar truss structure. In this, the method successfully calculated optimal design solutions that minimized the objective function while satisfying probabilistic constraints. This proves that the developed framework is theoretically sound and works effectively in solving high-dimensional nonlinear engineering design problems.

However, one important limitation was the instability in training the neural network part of the score matching algorithm. In some cases, it was observed that the optimization of the score network may be unstable, which could affect the accuracy of the score estimate. The improvement of this stability for robust convergence in a wide range of datasets and architectures is an important task for future studies.

References

- N. Wiener, "The Homogeneous Chaos," *Am. J. Math.*, Vol.60, No.4, pp. 897-936, 1938.
- D. Xiu and G. E. Karniadakis, "The Wiener-Askey Polynomial Chaos for Stochastic Differential Equations," *SIAM J. Sci. Comput.*, Vol.24, No.2, pp. 619-644, 2002.
- D. Lee and S. Rahman, "Robust design optimization under dependent random variables by a generalized polynomial chaos expansion," *Struct. Multidiscip. Optim.*, Vol.63, No.5, pp.2425-2457, 2021.
- Y. Song, S. Garg, J. Shi and S. Ermon, "Sliced Score Matching: A Scalable Approach to Density and Score Estimation," *Uncertainty in Artificial Intelligence (UAI)*, pp. 574-584, 2019.
- J. Ho, A. Jain and P. Abbeel, "Denoising Diffusion Probabilistic Models," *Advances in Neural Information Processing Systems*, Vol.33, pp.6840-6851, 2020.
- Y. Song, "Sliced Score Matching: A Scalable Approach to Density and Score Estimation," <https://yang-song.net/blog/2019/ssm/>